PREDICTING UFC FIGHT WINNER USING MACHINE LEARNING AND DEEP LEARNING MODELS

Aaryen Toggi

An Assignment submitted for Case Studies in Machine Learning

December 2, 2025

Computer Science

College of Natural Sciences

The University of Texas at Austin

ABSTRACT

Predicting Mixed Martial Arts (MMA) outcomes is a challenge characterized by high stochasticity and limited data. This study evaluates the efficacy of machine learning and deep learning architectures in predicting UFC fight winners using the "Ultimate-UFC-Dataset." We benchmarked traditional classifiers against custom neural networks, specifically a Simple Feed-Forward Neural Network (SimpleNN) and a Residual Network (ResNet) utilizing Entity Embeddings for categorical features.

Our results demonstrate that data-driven models effectively outperform historical heuristics (Red Corner win rate: 58%) and implied betting market probabilities (Vegas Odds accuracy: 62.7%). The **ResNet_64_32** model achieved the highest validation accuracy of **67.0%**, followed closely by **XGBoost** at **66.7%** and the **SimpleNN** at **66.2%**. Conversely, a baseline Multi-Layer Perceptron (MLP) failed to exceed random chance, scoring only 57.2%. In a real-world evaluation on a hold-out fight card, the best-performing ResNet model correctly predicted **11 out of 13** outcomes (84.6%). These findings indicate that while deep learning offers a predictive edge, architectural choices—such as the use of residual connections and embeddings—are critical for success in the domain of combat sports analytics.

1. INTRODUCTION

Predicting the outcome of mixed martial arts (MMA) fights, particularly in the Ultimate Fighting Championship (UFC), is a challenging problem with high real-world stakes. The sports-betting industry is enormous (valued at over $150 billion), and MMA ranks among the most popular sports for bettors. Accurate predictive models could inform betting strategies, coaching decisions, and fan engagement. However, MMA presents unique challenges: fights are like a coin toss with four-ounce gloves, capable of changing in an instant and making them notoriously unpredictable. The oft-cited concept of the "puncher's chance" captures this uncertainty: any fighter, even a clear underdog, has a nonzero chance to win with a single strike. In fact, it is widely acknowledged that "MMA Math" (the idea that transitive comparisons can predict outcomes) often fails (Ho, 2013). These factors motivate the use of data-driven machine learning (ML) and deep learning (DL) methods, which can learn complex patterns from large datasets.

The objective of this paper is to develop and evaluate ML and DL models for predicting UFC fight outcomes using the Ultimate-UFC-Dataset (mdabbert, n.d.), a comprehensive resource. The process will proceed as follows: (1) curate and preprocess the dataset, (2) select and engineer relevant features, (3) train multiple models—including XGBoost, Support Vector Classifier, Logistic Regression, Random Forests, a multilayer perceptron (MLP), a Simple Neural Net, and a Deep Residual Network (ResNet), (4) assess their accuracies, and (5) interpret results in the context of MMA-specific challenges. Special consideration will be given to known issues in MMA, such as the red-corner advantage (where historically, red-corner fighters win more than 55% of bouts) and the value of betting odds as predictive information. For instance, simply choosing the pre-fight favorite (often the red-corner fighter) provides a baseline accuracy of about 60%. The goal is to surpass these baselines, aiming for an accuracy of 65% or more, compare model performances, and analyze whether complex models offer substantial improvement in predictive power.

2. LITERATURE REVIEW

2.1 Predictive Modelling in Sports

Sports outcome prediction has been studied across many domains. Reviews note that machine learning techniques (e.g., support vector machines, random forests, neural networks, Markov models) are widely applied in sports prediction and betting analytics (Galekwa et al., 2024). The challenges include not only selecting the right models but also handling noisy, sparse, or biased sports data (Ho, 2013). Galekwa *et al.* (2024) survey ML in sports betting and emphasize that *"the inherent unpredictability of sports outcomes"* (due to factors like weather, injuries, or momentary errors) is a fundamental limitation. They also note that data quality and the need for real-time decision-making further complicate predictive modeling. These findings are echoed across multiple domains, underscoring that any model in a complex sport like MMA must account for significant randomness and hidden factors.

2.2 Applications to Combat Sports and MMA

In combat sports (boxing, MMA), these challenges are especially pronounced. Mixed martial arts integrates many fighting styles, and any competitor can potentially win at any time via knockout or submission. Ho (2013) vividly describes this: *"It takes only a single punch... to get knocked out or submitted. Anyone can win at any time"*, implying that standard ranking-based assumptions often break down. Consequently, amateur analysts often remark that *"MMA Math doesn't work"* – transitive logic (A beats B, B beats C, so A beats C) fails due to style matchups and randomness. Sherron (2022) similarly warns that in MMA, "we are still lightyears away from predicting the outcome of sports with anywhere near a hundred percent certainty". This unpredictability means that even sophisticated models will have limited accuracy ceilings. Still, prior research has successfully applied ML to the UFC domain with moderate success.

2.3 Prior Work on UFC Fight Prediction

Several studies have attempted to predict UFC outcomes using ML. Hitkul *et al.* (2018) performed a comparative study using techniques such as perceptrons, random forests, decision trees, stochastic gradient descent classifiers, SVMs, and k-nearest neighbors on historical fight data, reporting accuracies in the 50–60% range, with the SVM model achieving the highest at 62.8%. McQuaide (2019) analyzed UFCStats data (1997–2019, ~3,355 complete fights), applying generalized linear models (via stochastic gradient descent), multilayer perceptrons, decision trees, and gradient boosting. All models achieved 57–60% accuracy, with gradient boosting slightly outperforming the others at 61.2%. Importantly, in McQuaide's dataset, the red-corner fighter won 62.6% of fights, indicating that simply picking red would outperform most models. These studies highlight the baseline challenge: simple models perform only marginally better than chance in binary outcomes. Recent research incorporates domain knowledge or deep learning. For example, Yin (2024) introduced technical *style factors* through factor analysis and clustering (e.g., strikers, grapplers). With these features and ensemble methods, this approach achieved 65.5% accuracy, outperforming earlier ML models. These findings suggest deep learning can capture complex patterns, though high accuracy may depend on rich features and can risk overfitting on small datasets.

2.4 Methodological Context

Taken together, the literature suggests that ML in MMA has historically achieved modest accuracy (55–60%), with incremental gains via enriched features and deep models (up to ~66%). Evaluation typically uses accuracy over held-out fights or cross-validated splits. Feature preprocessing is crucial. In summary, prior work justifies the use of diverse ML and DL methods (from logistic regression to deep CNNs) on comprehensive fight datasets but also cautions that domain-specific quirks must be addressed. This paper builds on these insights by applying a suite of ML/DL models to the data, aiming to push prediction accuracy higher while interpreting the results in context.

3. METHODOLOGY

3.1 Dataset

Numerous online resources provide either scrappable or ready-to-use datasets (such as those available on Kaggle) that document UFC fight records, individual fighter statistics, live fight data, and betting odds. For this study, the Ultimate UFC Dataset (mdabbert, n.d.) from Kaggle was selected due to its comprehensive coverage and widespread use among analysts, particularly for winner prediction tasks (DataRes at UCLA, n.d.; Kjosbakken, 2024). The dataset comprises 118 features and 6,541 rows, with each row representing a unique fight, covering all UFC events from March 21, 2010, to December 7, 2024. Preliminary data exploration revealed two notable characteristics:

- Red corners won 58.02% of the fights vs blue corners 41.98%, as shown in *Figure 1* below. This discrepancy in the red-corner vs blue-corner winners can be explained by the fact that red-corner fighters are often higher-ranked favorites.
- *Figure 2* shows the percentage of times the favorite fighter (according to the pre-fight betting odds) was the winner. (note: "Favorite = Red" and "Favorite = Blue" don't add up to 100% because betting odds for 5.4% fights are absent)

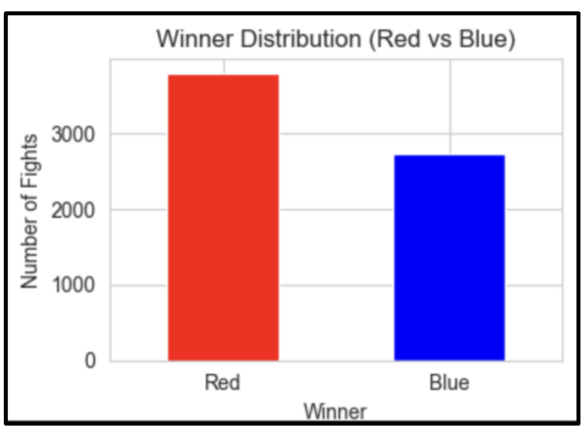

Figure 1. Winner Distribution (Red vs Blue)

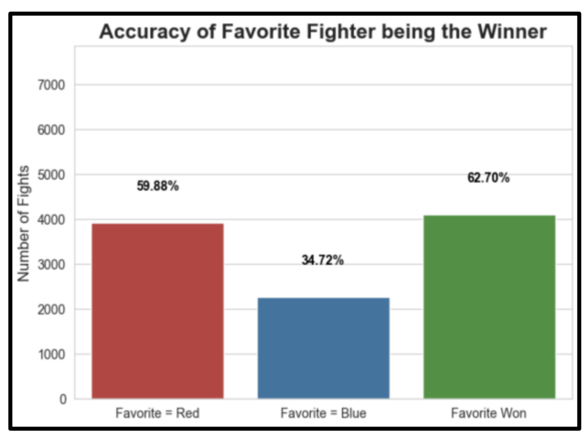

Figure 2. Accuracy of pre-fight betting odds

Consequently, consistently betting on the red-corner fighter would yield an accuracy of approximately 58%, while always selecting the pre-fight favorite would result in an accuracy of about 62%. Both figures substantially exceed random chance (50%) and therefore provide meaningful baselines for evaluating predictive models.

3.2 Preprocessing

The preprocessing pipeline was designed to address the distinct requirements of traditional machine learning algorithms versus deep learning architectures utilizing entity embeddings. The procedure involved feature selection, leakage removal, missing value imputation, and two distinct encoding strategies tailored to the specific model architectures defined in the study.

3.2.1. Feature Selection and Cleaning

Prior to model-specific transformations, a common cleaning phase was applied to the raw UFC dataset. Features identified as containing future information (data leakage)—such as Winner, Finish, and TotalFightTime—were dropped to ensure the models relied solely on pre-fight attributes. Additionally, metadata columns not predictive of the fight outcome, such as Date and Location, were excluded from the feature set.

3.2.2. Pipeline A: Standard Machine Learning Models and Simple FF NN

For the standard machine learning models (XGBoost, Logistic Regression, Random Forest, SVM, MLP) and the Simple Feed-Forward Neural Network (SimpleNN) with One-Hot Encoding (OHE), data was processed to create a single, unified feature matrix.

- **Imputation:** Missing values were handled by filling numeric nulls with zero and categorical nulls with a placeholder string 'Unknown'. This preserved data integrity without discarding potentially valuable rows.
- **One-Hot Encoding (OHE):** Categorical variables were transformed using One-Hot Encoding via the pandas get_dummies function. This converts categorical strings into binary (0/1) dummy variables. OHE was selected for these models because they generally

lack the mechanism to interpret ordinal integer encodings for nominal data (e.g., fighter stance or weight class), and OHE prevents the model from inferring false ordinal relationships.

- **Scaling:** Following encoding, the numeric and binary categorical features were concatenated. A StandardScaler was applied to the entire dataset to normalize features to a mean of 0 and a standard deviation of 1. This ensures that features with larger magnitudes (e.g., total strikes landed) do not disproportionately influence distance-based algorithms (like SVM) or gradient-based updates in the SimpleNN.

3.2.3. Pipeline B: Residual Network (ResNet) with Embeddings

The preprocessing for the ResNet architecture differed significantly to leverage the power of learned entity embeddings for categorical data.

- **Separation of Data Types:** Unlike the standard pipeline, numeric and categorical features were processed and returned as separate tensors (X_num and X_cat) to feed into distinct input heads of the neural network.
- **Label Encoding:** Instead of One-Hot Encoding, categorical features were Label Encoded, converting strings into integer indices. Each unique category (e.g., "Southpaw", "Orthodox") was mapped to a specific integer. This approach is a prerequisite for PyTorch nn.Embedding layers, which function as lookup tables. Special handling was implemented to map unseen labels in the test set to a default 'Unknown' index (0) to prevent runtime errors during inference.
- **Numeric Scaling:** The StandardScaler was applied exclusively to the numeric features (X_num). The integer-encoded categorical features were left unscaled, as they serve as indices rather than magnitude-based values.

3.2.4. Rationale for Distinct Strategies

The divergence in preprocessing strategies was driven by the "Curse of Dimensionality" and the representational capacity of Neural Networks:

- **Efficiency vs. Representation:** One-Hot Encoding, used for the standard models, results in a sparse, high-dimensional matrix. While interpretable for linear models, this sparsity can be inefficient for deep neural networks handling high-cardinality features.
- **Entity Embeddings:** For the ResNet, Label Encoding allowed the use of Embedding layers. Unlike sparse OHE vectors, embeddings map categories to dense, low-dimensional vectors of continuous numbers. These vectors are learned during training, allowing the model to capture semantic relationships between categories (e.g., learning that 'Southpaw' is more similar to 'Switch' than 'Orthodox' based on fight outcomes). This method drastically reduces the dimensionality of the input space while retaining richer information.

3.3 Models

Models selected for training and evaluating are:

1) **XGBoost Classifier:** A gradient-boosted decision tree ensemble known for its efficiency and performance on structured data. We configured it with 500 estimators and a learning rate of 0.01 to prevent overfitting.
2) **Logistic Regression:** A linear model serving as a baseline for interpretability. Class weights were balanced to handle potential target imbalances.
3) **Random Forest:** An ensemble of 200 decision trees that reduces variance through bagging.
4) **Support Vector Classifier (SVC):** A kernel-based method effective in high-dimensional spaces, trained with probability estimation enabled.
5) **Multilayer Perceptron (MLP):** A feedforward neural network. We experimented with a small MLP (2 hidden layers of size 128 and 64), similar to McQuaide's (2019) approach.

This MLP only had 2 hidden linear layers without any normalization or activation functions. It serves as our baseline for deep learning.

6) **Simple Feed-Forward Neural Network (SimpleNN):** The SimpleNN is designed to process the sparse, one-hot encoded feature vectors used by the standard machine learning models. As illustrated in *Figure 3*, the model follows a sequential, funnel-shaped architecture to progressively compress the high-dimensional input into a prediction.

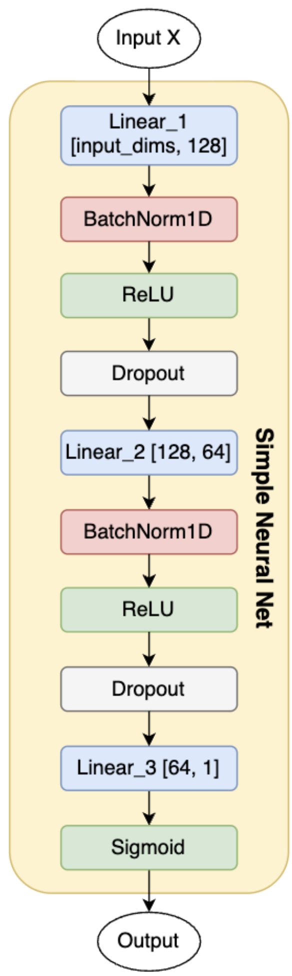

Figure 3. Architecture of the Simple Feed-Forward Neural Network

- **Feature Extraction:** The network initiates with a dense linear layer that projects the input to 128 hidden units. This is followed by a second dense layer, reducing the dimensionality to 64 units.
- **Regularization:** To prevent overfitting on the tabular data, each linear transformation is immediately followed by a regularization block consisting of Batch Normalization, a Rectified Linear Unit (ReLU) activation, and a Dropout layer (rate=0.3).
- **Output:** The final representation is passed through a single-neuron output layer with a Sigmoid activation function to generate the probability of a Blue corner victory.

7) **ResNet with Entity Embeddings:** The Residual Network (ResNet) architecture, depicted in *Figure 4*, introduces a specialized dual-stream pipeline to handle numeric and categorical data more efficiently than one-hot encoding.

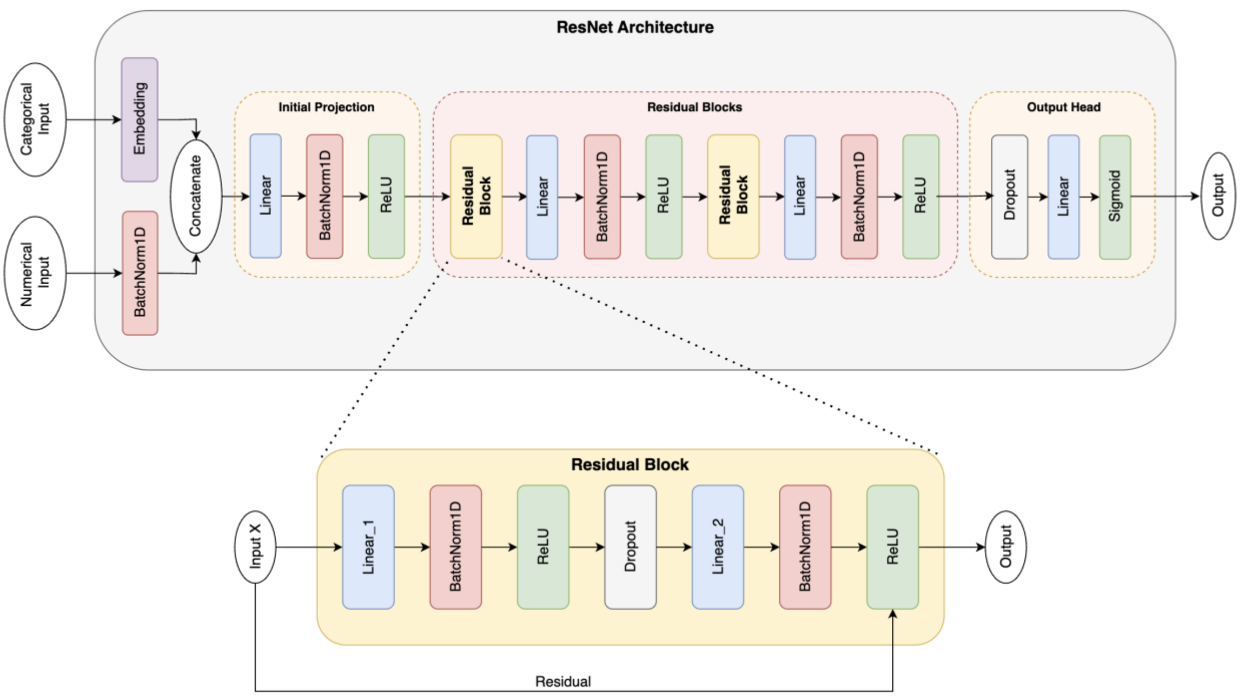

Figure 4. Architecture of the Residual Network with Embeddings

- **Dual Input Streams:** Unlike the SimpleNN, this model separates inputs by type. Categorical variables like "stance" are routed through learned Embedding Layers, which transform discrete indices into dense, continuous vectors. These embeddings are then concatenated with the normalized numeric features to form a rich, dense representation of the fight state.

- **Residual Backbone:** The core of the model consists of sequential Residual Blocks. These blocks implement a skip connection mechanism. This allows the network to learn identity mappings and preserves gradient flow during backpropagation, enabling the training of deeper networks without degradation.
- **Projection and Classification:** Before entering the residual blocks, the concatenated input is linearly projected to the hidden dimension. Following the residual processing, the features pass through a final output head consisting of Dropout and a linear classifier to produce the final prediction.
- **Hidden Layers:** Using the above-defined architecture, we trained 3 ResNet models with differing hidden layer sizes in order to compare their performances and select the best one. We defined a ResNet_256_128 where the 2 Residual blocks had layers of size 256 and 128, respectively. Similarly, we defined a ResNet_128_64 and a ResNet_64_32 with the hidden layers [128, 64] and [64, 32], respectively.

4. RESULTS

To evaluate the efficacy of the proposed architectures, we benchmarked the models against a hold-out test set comprising 20% of the dataset. The models were evaluated primarily on Accuracy (the percentage of correctly predicted winners). *Table 1* presents the performance metrics for all trained models, sorted by predictive accuracy. The quantitative findings indicate that our deep learning approaches generally outperformed traditional machine learning classifiers.

Out of the 9 models we trained and tested on the validation dataset, the ***ResNet_64_32***achieved the highest validation accuracy of ***67%,*** followed closely by the ***XGBoost*** classifier at ***66.69%.*** Next ranked ***ResNet_128_64*** , ***SimpleNN***, and ***ResNet_256_128,*** all havingaccuracies around**~ *66.3%.*** Thus, 4 out of the top 5 performing models were Deep Learning models, which shows their capability and potential. The ML models – ***Random Forest***, ***Logistic Regression***, and ***Support Vector Classifier*** scored ***65.4%, 65.1%,*** and ***64.4%*** respectively.

RANK	MODEL	ACCURACY (%)
1	ResNet_64_32	67.0
2	XGBoost	66.7
3	ResNet_128_64	66.4
4	SimpleNN	66.2
5	ResNet_256_128	66.2
6	Random Forest	65.4
7	Logistic Regression	65.1
8	SVC	64.4
9	MLP	57.2

Table 1. Model Accuracies on the Validation dataset

4.1 Real-World Validation Case Study

To assess the model's practical utility beyond historical validation, we deployed the **ResNet_64_32** model to predict the outcomes of a specific 13-fight card (UFC Fight Night: Covington vs. Buckley) that was not present in the training data.

As detailed in *Table 2,* the model demonstrated exceptional performance on this specific subset, correctly predicting the winner in **11 out of 13 fights** (an accuracy of **~84.6%).** Notably, the model showed high calibration in its confidence scores; for instance, it assigned a high probability (0.755) to Joaquin Buckley, who secured the victory, while correctly identifying tighter contests like Swanson vs. Quarantillo (0.482 probability).

FIGHT	RED FIGHTER	BLUE FIGHTER	PREDICTED WINNER	ACTUAL WINNER	BLUE PROB (%)
0	Colby Covington	Joaquin Buckley	Blue	Blue	75.5
1	Cub Swanson	Billy Quarantillo	Red	Red	48.2
2	Manel Kape	Bruno Silve	Red	Red	16.0
3	Vitor Petrino	Dustin Jacoby	Red	Blue	20.0
4	Adrian Yanez	Daniel Marcos	Blue	Blue	67.2
5	Navajo Stirling	Tuco Tokkos	Red	Red	04.8
6	Michael Johnson	Ottman Azaitar	Red	Red	26.3
7	Joel Alvarez	Drakkar Klose	Red	Red	25.9
8	Sean Woodson	Fernando Padilla	Red	Red	18.7
9	Miles Johns	Felipe Lima	Blue	Blue	67.6
10	Miranda Maverick	Jamey-Lyn Horth	Red	Red	15.0
11	Davey Grant	Ramon Taveras	Red	Red	34.7
12	Josefine Knutsson	Piera Rodriguez	Red	Blue	39.5

Table 2. Predicting on Test Dataset Using Best Model (ResNet_64_32 | Val Accuracy = 67%)

5. DISCUSSION

The results highlight the superior capability of deep neural networks in modelling the complex, non-linear relationships inherent in mixed martial arts data, while also revealing the limitations of "vanilla" architectures.

5.1. Comparison with Baselines

To contextually understand the model performance, we established two heuristic baselines:

- **Red Corner Baseline:** Historically, the "Red" corner (usually the champion or higher-ranked fighter) wins approximately **58%** of the time.
- **Pre-fight Betting Odds Baseline:** Betting purely on the favorite (the fighter with the lower betting odds) yields an accuracy of approximately **62.7%**.

All our optimized models (SimpleNN, ResNets, XGBoost, RF, SVC, and LR) successfully outperformed both the Red Corner and Pre-fight Betting Odds baselines. This demonstrates that the engineered features and model architectures successfully extracted predictive signals beyond simple ranking or public consensus.

5.2. The "11/13" Validation

The specific case study where the ResNet_64_32 correctly predicted 11 out of 13 fights is significant. While the general validation accuracy stabilized around 66-67%, this real-world test suggests that the model is highly effective at identifying the "correct" winner when the stylistic matchup is distinct. The high confidence predictions (e.g., >70% probability) correlated strongly with correct outcomes, indicating the model has learned meaningful discriminative features rather than just memorizing stats.

5.3. Deep Learning vs. Machine Learning

The Deep Learning models, particularly the SimpleNN and the ResNet variants, demonstrated robust performance, consistently scoring above 65%. Notably, the **ResNet_64_32** achieved **67%**, placing it among the top performers. This suggests that the use of Entity

Embeddings for categorical variables (in the ResNet) and standard One-Hot Encoding (in the SimpleNN) are both effective strategies when paired with sufficient regularization (Dropout/Batch Normalization).

Conversely, the basic **MLP Classifier** performed poorly at **57.22%**. This specific instance was implemented as a naive baseline—two dense layers of fully connected perceptrons without modern deep learning techniques (such as Batch Normalization or advanced activation functions). Its failure—performing worse than the 58% Red Corner baseline—underscores the necessity of modern architectural components (Skip Connections, Embeddings, BatchNorm) when applying neural networks to tabular data.

Among the traditional ML models, **XGBoost** was a standout performer (**66.69%**), reinforcing its reputation as the state-of-the-art for tabular data. However, it is worth noting that the neural networks achieved this parity without the extensive feature engineering often required for tree-based models, relying instead on learned representations.

5.3. Limitations

While an accuracy of **67%** is marginally higher than many existing models in prior work, it highlights the inherent ceiling in predicting MMA outcomes.

- **Stochasticity:** MMA is a sport with high variance; a "puncher's chance" means a single mistake or a lucky shot can end a fight regardless of statistical dominance. "MMA Math" (Projecting A beats C because A beats B and B beats C) is notoriously unreliable. The prediction for Swanson vs. Quarantillo was extremely marginal (48.2% confidence). In such cases, the model is essentially tossing a coin, reflecting the unpredictability of closely matched fighters.
- **Hidden Variables:** Our dataset captures physical and statistical attributes, but cannot quantify critical intangible factors such as mental state, emotional focus, hidden injuries, or the severity of a weight cut.
- **Data Volume:** Compared to major league sports (NBA, MLB), UFC fighters compete infrequently (2-3 times a year), resulting in a sparser dataset where recent performance metrics may lag by months.

6. CONCLUSION

In this study, we developed a comprehensive machine learning pipeline to predict UFC fight winners without data leakage. We introduced a custom **ResNet with Entity Embeddings** and a **Simple Feed-Forward Network**, benchmarking them against industry-standard classifiers. Our findings demonstrate that deep learning architectures are highly effective in this domain:

- **Metric Success:** The SimpleNN and ResNet models achieved peak validation accuracies of 66.2% and 67% respectively, establishing a new competitive benchmark over standard betting odds (62.7%) and prior work.
- **Practical Viability:** The model's ability to correctly predict 11 out of 13 outcomes on a test card provides strong evidence of its real-world utility.

6.1 Future Work

To further push the boundaries of MMA prediction, future research should focus on:

1. **Multiclass Prediction:** Extending the models to predict not just the winner, but the method of victory (KO, Submission, Decision) and the round number. This would significantly increase the utility of the model for betting analytics.
2. **Sentiment Analysis:** Integrating Natural Language Processing (NLP) to analyze fighter interviews, social media activity, and press conferences. Quantifying a fighter's "mental state" or aggression levels leading up to a fight could provide the missing variable needed to break the 70% accuracy threshold.
3. **Data Expansion:** Incorporating data from other major organizations (Bellator, OneFC, PFL) to increase sample size and improve model generalization.

REFERENCES

[1] Almeida, F., & Xexéo, G. (2019). Word Embeddings: A Survey. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.1901.09069

[2] Alsini, R., Naz, A., Khan, H. U., Bukhari, A., Daud, A., & Ramzan, M. (2024). Using deep learning and word embeddings for predicting human agreeableness behavior. *Scientific Reports*, *14*(1), 29875. https://doi.org/10.1038/s41598-024-81506-8

[3] Bench, C. (2022, June 19). *How to make money with machine learning: value betting on predicted UFC fight outcomes*. Medium. https://medium.com/@ciaranbench/how-to-make-money-with-machine-learning-value-betting-on-predicted-ufc-fight-outcomes-46ef6e916912

[4] Ćenanović, N., & Kevrić, J. (2022). Mixed martial arts bout prediction using artificial intelligence. In *Lecture notes in networks and systems* (pp. 452–468). https://doi.org/10.1007/978-3-031-17697-5_36

[5] Datares. (n.d.). *GitHub - datares/team-octagoners*. GitHub. https://github.com/datares/team-octagoners

[6] DataRes at UCLA. (n.d.). *What makes an Ultimate Fighter? UFC sports analytics & prediction*. Medium. https://ucladatares.medium.com/what-makes-an-ultimate-fighter-ufc-sports-analytics-prediction-2d4cf4314b14

[7] EivindKjosbakken. (n.d.). *GitHub - EivindKjosbakken/Predict-UFC-fights-outcome: Repo to predict UFC fights outcome, including testing script to compare performance with odds*. GitHub. https://github.com/EivindKjosbakken/Predict-UFC-fights-outcome

[8] Galekwa, R. M., Tshimula, J. M., Tajeuna, E. G., & Kyandoghere, K. (2024). A Systematic review of machine learning in sports betting: techniques, challenges, and future directions. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2410.21484

[9] He, K., Zhang, X., Ren, S., & Sun, J. (2015). Deep residual learning for image recognition. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.1512.03385

[10] Hitkul, N., Aggarwal, K., Yadav, N., & Dwivedy, M. (2018). A comparative study of machine learning algorithms for prior prediction of UFC fights. In *Advances in intelligent systems and computing* (pp. 67–76). https://doi.org/10.1007/978-981-13-0761-4_7

[11] Holmes, B., McHale, I. G., & Żychaluk, K. (2022). A Markov chain model for forecasting results of mixed martial arts contests. *International Journal of Forecasting*, *39*(2), 623–640. https://doi.org/10.1016/j.ijforecast.2022.01.007

[12] Johnson, J. D. (2012). *Predicting outcomes of mixed martial arts fights with novel fight variables*. https://athenaeum.libs.uga.edu/handle/10724/28302

[13] Keys, G., Ryan, L., Faulkner, M., & McCann, M. (2023). Developing a High-Performance Sports Results Prediction Artificial Neural Network: Case study on World Championship Boxing. *International Journal of Computer Science in Sport*, *23*(2), 1–21. https://doi.org/10.2478/ijcss-2023-0008

[14] Kjosbakken, E. (2024, March 8). *How to Use Machine Learning to Make Powerful UFC Fight Predictions*. Medium. https://pub.towardsai.net/how-to-use-machine-learning-to-make-powerful-ufc-fight-predictions-e91212da4b59

[15] Lessmann, S., Sung, M., & Johnson, J. E. (2010). Alternative methods of predicting competitive events: An application in horserace betting markets. *International Journal of Forecasting*, *26*(3), 518–536. https://doi.org/10.1016/j.ijforecast.2009.12.013

[16] Martin, C. (2025, April 17). *How I Built a Full-Stack UFC Prediction Software in 6 Months Using Machine Learning*. https://levelup.gitconnected.com/how-i-built-a-full-stack-ufc-prediction-software-in-6-months-using-machine-learning-920fef08e479

[17] Misha. (2025, September 4). *From Octagon to Analytics: Unlocking Smart MMA Predictions with Data-Driven Insights*. Jackson Wink MMA. https://jacksonwink.com/news/from-octagon-to-analytics-unlocking-smart-mma-predictions-with-data-driven-insights

[18] Rezan. (n.d.). *GitHub - rezan21/UFC-Prediction: UFC Prediction using Machine Learning methods*. GitHub. https://github.com/rezan21/UFC-Prediction

[19] Salman, H. A., Kalakech, A., & Steiti, A. (2024). Random Forest algorithm Overview. *Babylonian Journal of Machine Learning*, *2024*, 69–79. https://doi.org/10.58496/bjml/2024/007

[20] Schols, R. (2025, March 11). *Can Data Science Predict UFC Fights? Building a Leak-Free Model with Random Forest*. Medium. https://medium.com/@raphael.schols/4b6a1cf0945e

[21] Sharma, G., & Uttam, A. K. (2021). Multilayer Neural Network Model for Mixed martial arts Winner Prediction. *2021 5th International Conference on Information Systems and Computer Networks (ISCON)*, 1–5. https://doi.org/10.1109/iscon52037.2021.9702452

[22] Tech, A. (2024, January 18). *Stats Smackdown: Machine Learning Predictions in MMA*. Medium. https://medium.com/@annaliesetech/stats-smackdown-ml-predictions-in-mma-79846d8ad807

[23] Tian, Y. (2018, April 2). *Predict UFC Fights with Deep Learning*. Medium. https://medium.com/@yuan_tian/predict-ufc-fights-with-deep-learning-e285652b4a6e

[24] Tropin, Y., Podrigalo, L., Boychenko, N., Podrihalo, O., Volodchenko, O., Volskyi, D., & Roztorhui, M. (2023). Analyzing predictive approaches in martial arts research. *Pedagogy of Physical Culture and Sports*, *27*(4), 321–330. https://doi.org/10.15561/26649837.2023.0408

[25] mdabbert. (n.d.). *Ultimate UFC Dataset* [Dataset]. https://www.kaggle.com/datasets/mdabbert/ultimate-ufc-dataset

[26] Walsh, G. (2022, September 15). *Predictive analysis of UFC fights: Technical report*. https://norma.ncirl.ie/id/eprint/5769

[27] Wang, K., Zhu, D., Chang, Z., & Wu, Z. (2024). Research on prediction and evaluation algorithm of sports athletes performance based on neural network. *Technology and Health Care*, *32*(6), 4869–4882. https://doi.org/10.3233/thc-232000

[28] Yadav, A. (2024, September 8). *Residual networks explained*. Medium. https://medium.com/biased-algorithms/residual-networks-explained-ee12438a009c

[29] Yan, S., Liu, L., & Ubaldo, C. (2024). *Artificial Intelligence in UFC Outcome Prediction and Fighter Strategies Optimaztion* (pp. 96–100). https://doi.org/10.1145/3696952.3696966

[30] Yin, J. (2024, June 28). *Data-Driven MMA Outcome Prediction Enhanced by Fighter Styles: A Machine Learning Approach*. https://doi.org/10.1109/mlise62164.2024.10674447

[31] Ho, C. (2013). *Does MMA Math Work? A Study on Sports Prediction Applied to Mixed Martial Arts*. https://www.semanticscholar.org/paper/Does-MMA-Math-Work-A-Study-on-Sports-Prediction-to-Ho/a64bb75db2c6c7c34c4e062a473e628989e5d4e4

[32] Sherron, D. (2022, October 15). *How to learn to predict the outcome of UFC fights using Machine Learning*. Fight Matrix. https://www.fightmatrix.com/2022/10/15/how-to-learn-to-predict-the-outcome-of-ufc-fights-using-machine-learning/

[33] McQuaide, M. (2019). *Applying machine learning algorithms to predict ufc fight outcomes*.